Unit II

2. System Planning, Feasibility study & Cost-Benefit Analysis

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2.1 Basis for planning in system analysis: Dimensions of Planning:

Basis for Planning in System Analysis:

Planning in system analysis lays the foundation for successful system development. It involves identifying **what** needs to be done, **how**, and **when**, based on organizational goals and system requirements.

Dimensions of Planning:

1. Scope Planning:

Defines the boundaries of the system.
Identifies what will be included and excluded.

2. Resource Planning:

Determines the resources required — **people, time, budget, tools**, etc.
Ensures efficient allocation.

3. Time Planning (Scheduling):

Estimates the **timeframe** for each phase.
Helps in setting deadlines and milestones.

4. Cost Planning:

Involves **budget estimation** and financial planning.
Ensures the system remains cost-effective.

5. Risk Planning:

Identifies potential risks and prepares strategies to **avoid or handle** them.
Improves system reliability and success.

These dimensions ensure that the system development is **well-organized, efficient**, and meets user expectations.

2.2 Initial Investigation: Determining user's requirements and analysis:

Definition:

Initial investigation is the **first step** in system analysis, where the system analyst studies the existing system, understands user needs, and identifies the problem to be solved.

Steps Involved:

1. Understanding the Problem:

Identify why a new system is needed.
Understand the current issues or limitations.

2. Identifying Objectives:

Define what users expect from the new system.

Set clear goals based on user input.

3. Fact-Finding Techniques:

Use interviews, questionnaires, observations, and document reviews to gather information from users.

4. Analyzing User Requirements:

Study the collected data to understand **what users want**, not just **what they say**.

Identify both **functional** and **non-functional** requirements.

5. Preparing a Report:

Document findings and suggest whether to proceed with the full system study.

Importance:

It ensures that the project starts with a **clear understanding of user needs**, reducing errors and rework later.

2.3 fact finding process and techniques:

Definition:

The **fact-finding process** is the method used by system analysts to gather information about the existing system and user requirements. It helps in understanding the current problems and expectations for the new system.

Fact-Finding Techniques:

1. Interviewing:

Face-to-face interaction with users to gather detailed insights.

Helps in understanding opinions and expectations.

2. Questionnaires:

Set of written questions given to users.

Useful for collecting data from a large number of people quickly.

3. Observation:

Analyst watches users perform tasks to understand how the current system works.

Reveals real-world practices and issues.

4. Document Review:

Studying existing reports, forms, manuals, and system documents.

Helps understand procedures and historical data.

5. Record Sampling:

Examining a sample of past records or transactions.

Used for analyzing trends or common issues.

Importance:

Accurate fact-finding ensures the system is built based on **real needs**, improving its success and user satisfaction.

2.4 Feasibility study: Technical:

Definition:

Technical feasibility is a part of the overall feasibility study that evaluates whether the **technology and resources** needed to develop the proposed system are available and capable of meeting the requirements.

Key Aspects of Technical Feasibility:

1. Hardware and Software Availability:

Checks if the required hardware and software are available or need to be purchased or developed.

2. Technical Expertise:

Assesses whether the team has the necessary technical skills and experience to build and maintain the system.

3. System Compatibility:

Ensures that the new system will work smoothly with the existing systems and infrastructure.

4. Scalability and Performance:

Evaluates if the system can handle future growth in data and users efficiently.

5. Time Constraints:

Determines whether the system can be developed and implemented within the required timeframe using available technology.

Conclusion:

If a system is **technically feasible**, it means it is possible to build and run it using current technology, tools, and expertise.

2.5 Operational & Economic Feasibilities:

1. Operational Feasibility:

Definition:

Operational feasibility checks whether the proposed system will **function effectively in the user environment** and if it will be accepted by users.

Key Points:

User Acceptance: Will users be comfortable and willing to use the new system?
Work Environment Fit: Can the system be integrated into existing workflows?
Training Needs: Are users ready or do they need training to operate the system?
Support and Maintenance: Are there resources available to maintain and support the system?

Conclusion:

If a system is **operationally feasible**, it means it will work practically and meet user expectations in a real-world setting.

2. Economic Feasibility:

Definition:

Economic feasibility checks whether the **benefits** of the proposed system **outweigh the costs** involved in developing and implementing it.

Key Points:

Cost Analysis: Includes development, hardware, software, training, and maintenance costs.

Benefit Analysis: Looks at savings, efficiency, profit, or service improvements.

Cost-Benefit Comparison: Compares total costs with expected financial or non-financial benefits.

Conclusion:

If a system is **economically feasible**, it is considered a good investment for the organization.

2.6 Cost/Benefit Analysis:

Definition:

Cost/Benefit Analysis (CBA) is a technique used to **compare the costs** of developing and implementing a system with the **expected benefits**, to determine whether the project is financially worthwhile.

Types of Costs:

1. Development Costs:

Expenses for software, hardware, labor, training, etc.

2. Operating Costs:

Ongoing costs like maintenance, support, upgrades, etc.

Types of Benefits:

1. Tangible Benefits:

Measurable in money (e.g., reduced labor costs, increased sales).

2. Intangible Benefits:

Not directly measurable (e.g., improved customer satisfaction, better decision-making).

Purpose:

Helps in **decision-making** by showing if benefits outweigh costs.

Supports **economic feasibility** during system planning.

Conclusion:

If the **total benefits** are greater than the **total costs**, the system is considered **cost-effective and worth implementing**.